

Ayush Khot

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Education

University of Illinois at Urbana-Champaign, Urbana-Champaign, IL

August 2025 - May 2030

Doctor of Philosophy: COMPUTER SCIENCE

GPA: 3.91/4.00

Advisor: SHAOWEN WANG

University of Illinois at Urbana-Champaign, Urbana-Champaign, IL

August 2021 - May 2025

Bachelor of Science: COMPUTER SCIENCE

GPA: 3.96/4.00

Bachelor of Science: ENGINEERING PHYSICS

Experience

National Center for Supercomputing Applications, Urbana, IL

May 2022 - May 2025

UNDERGRADUATE RESEARCH ASSISTANT

- Examine, identify shortcomings with, and develop improved explainable AI (xAI) methods on deep neural networks in PyTorch and TensorFlow, reaching a state-of-the-art 93.2% accuracy in top tagging
- Aggregate critical physical characteristics pertinent to top quark decay using xAI metrics like SHAP and LRP
- Compare evidential deep learning (EDL) with baseline methods for uncertainty quantification and anomaly detection in jet tagging causing an increase in AUC by 0.05 in uncertainty quantification and variable improvements in anomaly detection
- Suggest new "Confidence Tuned" variance of EDL that has more accurate uncertainty quantification and similar accuracy
- Examine EDL uncertainties in latent space and the impact of pairwise particle interactions
- Write papers as first author on both applications of XAI methods and EDL on jet tagging (both accepted to MLST)

Brookhaven National Laboratory, Upton, NY

June 2024 - August 2024

MACHINE LEARNING GROUP INTERN

- Adopt evidential deep learning (EDL) into storm forecasting to estimate uncertainty in model predictions
- Achieve better calibrated uncertainty using EDL as compared to baseline methods like Ensemble and Monte Carlo Dropout while requiring 10 times less inference time to estimate uncertainties and sustain similar baseline MSE Loss using EDL (0.0044) as compared to baseline methods (0.0036)
- Presented a poster and oral presentation at Brookhaven National Laboratory, and presented a NeurIPS workshop paper as first author.

National Center for Supercomputing Applications, Urbana, IL

August 2023 - May 2024

UNDERGRADUATE RESEARCH INTERN

- Employ multi-gpu training for physics-informed neural operators (PINOs) to model 2D magnetohydrodynamics (MHD) simulations
- Contributed to NVIDIA's PhysicsNeMo package by implementing new PINO models for modeling MHD and nonlinear shallow water simulations

Selected Publications

Conference and Journal Publications

- [1] **A. Khot et al**, "Applying foundation model embeddings towards urban livability evaluation." 34th ACM SIGSPATIAL International Conference on Advances in Geographic Information Systems, Nov 2026.
- [2] **A. Khot et al**, "Spatial Deconfounder: Interference-Aware Deconfounding for Spatial Causal Inference." 43rd International Conference on Machine Learning (ICML), July 2026.
- [3] **A. Khot et al**, "Evidential deep learning for uncertainty quantification and out-of-distribution detection in jet identification using deep neural networks." Machine Learning: Science and Technology, July 2025.
- [4] **A. Khot et al**, "Evidential deep learning for probabilistic modelling of extreme storm events." Machine Learning and the Physical Sciences Workshop at the 38th conference on Neural Information Processing Systems (NeurIPS), Sep 2024.
- [5] **A. Khot et al**, "A detailed study of interpretability of deep neural network based top taggers." Machine Learning: Science and Technology, July 2023.

Talks

- **A. Khot**, W. Zuo, S. Wang, "Recovering Socioeconomic Information from AlphaEarth Embeddings." Presented at American Association of Geographers Annual Meeting (AAG), San Francisco, CA, Mar. 2026.
- **A. Khot**, X. Luo, A. Kagawa, "Evidential Deep Learning for Probabilistic Storm Detection." Presented at Brookhaven National Laboratory, Upton, NY, Aug. 2024.
- **A. Khot**, X. Wang, M. Neubauer, V. Kindratenko, A. Roy, "Uncertainty Quantification and Anomaly Detection with Evidential Deep Learning." Presentation at National Center for Supercomputing Applications: Students Pushing Innovation project, Urbana, IL, April 2024.

Honors & Awards

2025 - 2030	NSF Graduate Research Fellowship Program (GRFP) , National Science Foundation	<i>Urbana, IL</i>
2024	Philip J. and Betty M. Anthony Undergraduate Summer Research Scholarship , University of Illinois Department of Physics	<i>Urbana, IL</i>
2021 - 2024	Shelton M&R Matthews Scholarship , University of Illinois	<i>Urbana, IL</i>

Leadership Experience

NetMath at University of Illinois at Urbana-Champaign *Feb 2022 - August 2023*

MENTOR

- Facilitate weekly communication and oversee a group of over 20 Multivariable Calculus students
- Utilize Mathematica to meticulously grade homework assignments and deliver comprehensive feedback to students
- Conduct thorough reviews of exam results to ensure accurate assessment and continuous improvement in their understanding of the material

Illini Strings (UIUC Orchestra) *August 2021 - May 2025*

CORE MEMBER

Skills

Programming	Python, C/C++, JAVA, $\text{\LaTeX}$, Linux, Parallel Computing (CUDA, DDP, Horovod), Machine Learning (PyTorch, TF)
Relevant Courses	Num. Analysis, Machine Learning, Parallel Programming, Linear Algebra, Data Structures and Algorithms
HPC Experience	NERSC Perlmutter, NCSA Delta, NCSA HAL, NCSA TGI RAILS, ALCF Polaris